Solvency Field Theory

V10.1: Field-Theoretic Shatter, CMB Transfer, and Solver-Validated Low-ℓ Phenomenology

A standalone extension paper building on the V10 framework

This paper consolidates the full V10.1 extension of the early-universe shatter program into one standalone document. It incorporates the field-theoretic shatter mechanism, the independent-clock decoupling law, the minimal Boltzmann tie-in, the first-principles origin of the transfer coefficient β_Q, the instability-band selection rule, the first live CLASS implementation, the calibration against the pre-Boltzmann source builder, and the low-ℓ phenomenology closure. The intent is to preserve the V10.1 result as a single paper that is readable as a framework paper while still carrying the derivation spine and validation chain needed for later migration into a V11 main text plus a separate full-derivation archive.

April 17, 2026

Abstract

We extend Solvency Field Theory (SFT) beyond the V10 framework by deriving the early-universe "shatter" as a field-theoretic instability of the lag field, driven by a ledger solvency ratio

$$\Lambda(\mathbf{x}, t) = \frac{\dot{s}(\mathbf{x}, t)}{H\, i(\mathbf{x}, t)}.$$

The unstable phase occurs when $\Lambda > 1$, giving an effective mass

$$M_{\text{eff}}^2 = A(1 - \Lambda)$$

and a broken-phase order parameter

$$\Phi_{\text{eq}} \propto \sqrt{\Lambda - 1}.$$

We then connect the resulting relic phase field to the photon-baryon fluid through a minimal Boltzmann source term, show that the earlier thin-shell bridge is recovered as the narrow-source limit of a visibility-weighted transfer kernel, derive the first-principles origin of the coefficient $\beta_Q(\eta)$ and the k^2/a^2 source structure from the shatter free-energy functional, and obtain a band-selection rule

$$k < k_c(\eta) = \frac{a(\eta)}{c_\chi}\sqrt{A(\Lambda(\eta) - 1)}.$$

The full source was implemented in CLASS as an additive scalar temperature source. The live solver reproduces zero-source recovery exactly, validates the expected amplitude, transfer-factor, source-width, and band-position scalings, and confirms the C16 instability-band rule at solver level. After calibrating the analytic CLASS mode to the pre-Boltzmann source builder, the physical baseline produces a low-ℓ TT effect of about $+0.34$ to $+0.37\,\mu\text{K}^2$ at $\ell = 4$ with

$$\Delta\chi^2(\ell \leq 29) = 2 \times 10^{-6},$$

placing the source deeply inside the Planck-allowed region while preserving the acoustic peaks to machine precision. We conclude that the V10.1 shatter/CMB sector has reached a v1 phenomenology closure threshold: it is symbolically derived to the current frontier, numerically implemented, calibrated, solver-validated, and likelihood-mapped at low ℓ.

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1 Scope and contributions relative to V10

The V10 framework established the structural SFT backbone and the candidate early-universe projection story. V10.1 upgrades that early-universe sector in six concrete ways:

1. it derives a field-theoretic shatter instability for the lag field rather than treating the relic phase field as an inserted ansatz,
2. it identifies the control parameter as the independent-clock burden fraction of the maintenance ledger,
3. it ties the resulting relic field into the photon-baryon sector through a minimal Boltzmann source,
4. it derives the source coefficient $\beta_Q(\eta)$ and the k^2/a^2 coupling structure from the shatter free-energy functional,
5. it implements the source in the CLASS Boltzmann solver,
6. it calibrates and maps the resulting low-ℓ phenomenology against a cosmic-variance penalty metric.

The purpose of the present paper is therefore different from the V10 master paper. V10 introduced the framework. V10.1 is a standalone extension paper focused on one specific question: can the shatter mechanism be carried from ledger physics all the way to solver-real, calibrated, and observationally allowed CMB phenomenology?

Executive result. The answer is yes at the v1 phenomenology level. The shatter source is now: (i) field-theoretically derived to the current frontier, (ii) translated into Boltzmann-solver language, (iii) implemented and validated in CLASS, and (iv) shown to lie deeply within the Planck-allowed low-ℓ region at the physical calibrated normalization.

2 Field-theoretic shatter from ledger insolvency

2.1 Local solvency ratio and control parameter

The global SFT backbone is

$$H = \frac{\dot{S}}{I}. \tag{1}$$

To formulate a shatter, we promote this to a local control quantity by defining a maintenance demand density $\dot{s}(\mathbf{x}, t)$ and an accommodation density $i(\mathbf{x}, t)$. The local solvency ratio is

$$\Lambda(\mathbf{x}, t) = \frac{\dot{s}(\mathbf{x}, t)}{H\, i(\mathbf{x}, t)}. \tag{2}$$

The physical interpretation is immediate:

$$\begin{aligned} \Lambda < 1 &\quad \text{solvent / stable,} \\ \Lambda = 1 &\quad \text{critical,} \\ \Lambda > 1 &\quad \text{insolvent / unstable.} \end{aligned}$$

Define the corresponding control parameter

$$\epsilon(\mathbf{x}, t) = \Lambda(\mathbf{x}, t) - 1. \tag{3}$$

This is the SFT analog of a thermodynamic margin-call variable.

2.2 Minimal local action and corrected sign structure

Promote the lag variable to a local field,

$$\chi(t, \mathbf{x}) = \bar{\chi}(t) + \delta\chi(t, \mathbf{x}), \tag{4}$$

and write the mildest local effective action consistent with locality and the $\chi \to -\chi$ symmetry,

$$S_\chi = \int d^4x \sqrt{-g} \left[\frac{K}{2} g^{\mu\nu} \partial_\mu \chi \, \partial_\nu \chi - V(\chi, \Lambda) \right], \tag{5}$$

with the corrected potential

$$V(\chi, \Lambda) = \frac{A}{2}(1 - \Lambda)\chi^2 + \frac{\lambda}{4}\chi^4, \qquad A > 0, \ \lambda > 0. \tag{6}$$

This sign convention is load-bearing. It guarantees that the shared regime is stable when $\Lambda < 1$ and unstable when $\Lambda > 1$.

The corresponding effective mass is

$$M_{\text{eff}}^2 = A(1 - \Lambda), \tag{7}$$

so

$$\begin{aligned} \Lambda < 1 &\Rightarrow M_{\text{eff}}^2 > 0, \\ \Lambda = 1 &\Rightarrow M_{\text{eff}}^2 = 0, \\ \Lambda > 1 &\Rightarrow M_{\text{eff}}^2 < 0. \end{aligned}$$

On a flat FLRW background, the linear perturbation equation becomes

$$\ddot{\delta\chi} + 3H\dot{\delta\chi} - \frac{1}{a^2}\nabla^2 \delta\chi + A(1 - \Lambda)\, \delta\chi = 0. \tag{8}$$

This is the first honest field-theoretic shatter equation in SFT language.

2.3 Broken-phase order parameter

The equilibrium states follow from

$$\frac{\partial V}{\partial \chi} = A(1-\Lambda)\chi + \lambda\chi^3 = 0. \tag{9}$$

For the broken phase $\Lambda > 1$, the nonzero minima satisfy

$$\chi_{\rm eq}^2 = \frac{A}{\lambda}(\Lambda - 1), \tag{10}$$

so the post-critical order parameter is

$$\Phi_{\rm eq} \propto \sqrt{\Lambda - 1}. \tag{11}$$

At the minimal effective level, the relic phase field is therefore the broken-phase order parameter of the lag field rather than an inserted ansatz.

3 Independent-clock decoupling and the shatter control law

Write the total maintenance demand density as

$$\dot{s}_{\rm tot}(T) = \dot{s}_{\rm shared}(T) + \dot{s}_{\rm ind}(T), \tag{12}$$

where $\dot{s}_{\rm shared}$ is the maintenance load of the collective plasma regime and $\dot{s}_{\rm ind}$ is the load carried by emerging massive clocks that can no longer remain submerged in the shared bookkeeping system.

Define the independent-clock burden fraction

$$f(T) = \frac{\dot{s}_{\rm ind}(T)}{\dot{s}_{\rm shared}(T) + \dot{s}_{\rm ind}(T)}. \tag{13}$$

The key physical refinement is that the shatter is not controlled by raw thermal abundance alone. It is controlled by the fraction of the maintenance burden that can no longer remain shared.

The mildest effective decoupling law is

$$D(T) \approx \frac{1}{1 + (T/T_d)^p}, \tag{14}$$

with $D(T \gg T_d) \to 0$ and $D(T \ll T_d) \to 1$. At the effective level this gives

$$f(T) \approx \frac{1}{1 + (T/T_d)^p}, \tag{15}$$

and therefore the solvency history becomes

$$\Lambda(T) = \Lambda_0 + \Lambda_1 f(T). \tag{16}$$

The shatter threshold is then simply

$$\Lambda(T_c) = 1. \tag{17}$$

A bounded crossover scan showed that mild parameter choices place T_c naturally in the same thermal neighborhood as the earlier shatter-budget estimates, which is exactly the result one wants from a first effective decoupling law.

Status of this section. The field-theoretic instability and the order-parameter structure are derived at the minimal effective level. The exact first-principles derivation of the decoupling fraction $f(T)$ from a microscopic clock sector remains open, but the effective crossover law is now physically motivated rather than ad hoc.

4 Minimal Boltzmann tie-in and the narrow-source limit

4.1 Minimal source insertion

Define the photon temperature monopole by

$$\Theta_0 \equiv \frac{\delta_\gamma}{4}. \tag{18}$$

The minimal SFT heating source is taken to be proportional to the curvature of the relic phase field,

$$\delta Q_{\gamma,k}(\eta) = \beta_Q(\eta)\,\frac{k^2}{a(\eta)^2}\,\delta\Phi_k(\eta). \tag{19}$$

This is the mildest scalar source with the correct qualitative structure: it vanishes for $k = 0$, converts phase curvature into radiation response, and isolates the new physics in one conversion coefficient, $\beta_Q(\eta)$.

The photon continuity equation is then modified to

$$\dot{\Theta}_{0,k} = -\frac{k}{3}v_{\gamma,k} - \dot{\Phi}_{\mathrm{grav,k}} + \frac{\delta Q_{\gamma,k}}{4\rho_\gamma}. \tag{20}$$

In tight coupling, this feeds a sourced acoustic oscillator,

$$\ddot{\Theta}_{0,k} + \frac{\dot{R}}{1+R}\dot{\Theta}_{0,k} + c_s^2 k^2 \Theta_{0,k} = \mathcal{G}_k(\eta) + \mathcal{S}_{\mathrm{SFT},k}(\eta), \tag{21}$$

with

$$\mathcal{S}_{\mathrm{SFT},k}(\eta) \propto \frac{d}{d\eta}\left[\frac{\beta_Q(\eta)}{4\rho_\gamma(\eta)}\frac{k^2}{a(\eta)^2}\delta\Phi_k(\eta)\right]. \tag{22}$$

4.2 Connection to the shatter field and line-of-sight form

From the broken-phase result,

$$\delta\Phi_k(\eta) = \Phi_{\mathrm{eq}}(\eta)\,\eta_k, \tag{23}$$

so the line-of-sight source function can be written as

$$S_{\rm SFT}(k,\eta) \sim g(\eta)\,\frac{\beta_Q(\eta)}{4\rho_\gamma(\eta)}\,\frac{k^2}{a(\eta)^2}\,\delta\Phi_k(\eta), \tag{24}$$

where $g(\eta)$ is the visibility function. The observed multipoles then take the usual line-of-sight form,

$$\Theta_\ell^{\rm SFT}(k,\eta_0) = \int d\eta\, S_{\rm SFT}(k,\eta)\, j_\ell(k(\eta_0-\eta)). \tag{25}$$

In the narrow-source limit, with both the visibility and the heating block sharply localized near last scattering, this reduces to

$$\frac{\Delta T}{T} \sim K_{\rm cmb}\,\text{projection}[\nabla^2\delta\Phi], \tag{26}$$

so the older $K_{\rm cmb}$ bridge is recovered as the thin-shell reduction of a genuine transfer kernel rather than as a separate law.

A toy line-of-sight implementation confirmed exactly this pattern: broad-source histories looked strained, while an ultrathin, recombination-localized source gave a median full/thin ratio of 0.943811, validating the thin-shell picture in the appropriate limit.

5 First-principles origin of $\beta_Q(\eta)$ and the instability band

5.1 Shatter free energy and ledger chemical potential

The post-critical order parameter Φ is governed by the free-energy functional

$$F[\Phi] = \int d^3x\, a^3 \left[\frac{K_\Phi}{2a^2}(\nabla\Phi)^2 + \frac{A}{2}(1-\Lambda)\Phi^2 + \frac{\lambda}{4}\Phi^4\right]. \tag{27}$$

Its functional derivative is the ledger chemical potential

$$\mu_\Phi \equiv \frac{\delta F}{\delta\Phi} = -\frac{K_\Phi}{a^2}\nabla^2\Phi + A(1-\Lambda)\Phi + \lambda\Phi^3. \tag{28}$$

The photon-baryon fluid should couple to this thermodynamic drive rather than to Φ itself:

$$\delta Q_\gamma = \mathcal{C}_{\gamma\Phi}(\eta)\,\mu_\Phi. \tag{29}$$

5.2 Factorization of the coupling coefficient

The coupling coefficient decomposes into physically distinct pieces,

$$\mathcal{C}_{\gamma\Phi}(\eta) = q_T\, f_b(\eta)\, f_r(\eta)\, \epsilon_\Phi(\eta)\, \mathcal{R}_\Phi(\eta), \tag{30}$$

where

- $q_T = \frac{1}{4}$ is the radiation thermodynamic conversion factor,

- f_b is the fraction of the maintenance burden sourced by the baryonic / massive-clock sector,
- f_r is the fraction of released energy that is deposited into the radiation bath,
- ϵ_Φ is the phase-to-linear conversion factor,
- $\mathcal{R}_\Phi$ is the relaxation / transfer block.

Near the shatter, the dominant source piece is the gradient term in Eq. (28), so

$$\delta Q_\gamma \approx -q_T f_b f_r \epsilon_\Phi \mathcal{R}_\Phi K_\Phi \frac{\nabla^2 \delta\Phi}{a^2}. \tag{31}$$

Thus the source coefficient in Eq. (19) is

$$\beta_Q(\eta) = q_T\, f_b(\eta)\, f_r(\eta)\, \epsilon_\Phi(\eta)\, \mathcal{R}_\Phi(\eta)\, K_\Phi(\eta). \tag{32}$$

This is the first-principles origin of β_Q.

5.3 Why the source carries k^2/a^2

The k^2/a^2 dependence is not inserted by hand. It follows directly from the gradient term in Eq. (27). In Fourier space,

$$\nabla^2 \delta\Phi \to -k^2 \delta\Phi_k, \tag{33}$$

so

$$\delta Q_{\gamma,k} = \beta_Q(\eta)\, \frac{k^2}{a^2}\, \delta\Phi_k. \tag{34}$$

This is precisely the source structure used in the Boltzmann tie-in.

5.4 Candidate $4\pi^2$ splice and Hubble-limited closure

The natural SFT candidate remains

$$\epsilon_\Phi = \frac{1}{4\pi^2}. \tag{35}$$

If this holds, then the fifth appearance of $4\pi^2$ in the CMB sector is no longer a bare numerical coincidence but a phase-to-linear conversion factor inside Eq. (32). The symbolic theorem for Eq. (35) is not fully closed, but it is now embedded in a concrete source law.

The mildest closure for the remaining transport block is the Hubble-limited relaxation hypothesis,

$$\tau_\Phi \sim H^{-1}, \tag{36}$$

which gives

$$\beta_Q(\eta) \sim \frac{q_T\, f_b(\eta)\, f_r(\eta)\, \epsilon_\Phi(\eta)}{H(\eta)}. \tag{37}$$

This is the scale used later in the analytic CLASS mode.

5.5 Band-selected unstable spectrum

The linear lag-field perturbation equation implies that in the unstable phase $\Lambda > 1$, growth occurs only for modes satisfying

$$c_\chi^2 \frac{k^2}{a^2} < A(\Lambda - 1). \tag{38}$$

The unstable band is therefore

$$k < k_c(\eta) = \frac{a(\eta)}{c_\chi} \sqrt{A(\Lambda(\eta) - 1)}. \tag{39}$$

This is the C16 band-selection rule: the relic spectrum is a band-limited instability spectrum, not an unconstrained random Gaussian field.

6 CLASS implementation, calibration, and validation

6.1 Implemented source

The v1 CLASS implementation uses the additive scalar temperature source

$$S_T^{\rm new}(k,\tau) = S_T^{\rm std}(k,\tau) + S_{\rm SFT}(k,\tau), \tag{40}$$

with

$$S_{\rm SFT}(k,\tau) = A_{\rm sft}\, g(\tau)\, \beta_Q(\tau)\, \frac{k^2}{a(\tau)^2}\, \Phi_{\rm eq}(\tau)\, \eta_k\, W\!\left(\frac{k}{k_c(\tau)}\right). \tag{41}$$

The first implementation path modifies only:

- `include/perturbations.h`,
- `source/input.c`,
- `source/perturbations.c`,

with no required edits to `transfer.c` or the Python wrapper in v1.

6.2 Builder-to-patch calibration

The pre-Boltzmann builder and the analytic CLASS patch agree closely on the background shatter quantities at last scattering. Table 1 summarizes the key comparison.

Quantity	Builder value	Patch analytic value	Agreement
$\beta_Q(z_*)$	1.24×10^{-4}	2.85×10^{-4}	ratio = 2.30
$\Phi_{\rm eq}(z_*)$	0.4766	0.4737	0.6%
$k_c(z_*)$	$4.37 \times 10^{-4}\,{\rm Mpc}^{-1}$	$4.30 \times 10^{-4}\,{\rm Mpc}^{-1}$	1.6%

Table 1: Builder-to-analytic-patch calibration at last scattering.

The remaining mismatch is the expected missing relaxation block in the simplified analytic source. The mapped normalization is therefore

$$A_{\rm sft} = \frac{\beta_{Q,\rm builder}\,\Phi_{\rm eq,builder}}{\beta_{Q,\rm patch}\,\Phi_{\rm eq,patch}} \approx 0.4378. \tag{42}$$

This is not treated as a free fit parameter. In v1 it is the physical calibration ratio between the simplified analytic source and the fuller pre-Boltzmann source history.

6.3 Validation ladder

The live solver validation returned the expected sequence:

1. zero-source recovery is bit-identical to vanilla CLASS,
2. zero-amplitude recovery is also bit-identical,
3. the amplitude ladder is perfectly linear in $A_{\rm sft}$,
4. the ϵ_Φ ladder is nearly exactly linear, confirming a true multiplicative solver parameter,
5. the instability-band sweeps move the signal in ℓ-space exactly as Eq. (39) predicts,
6. visibility-width sweeps reproduce the thin-shell story found in the toy line-of-sight note.

7 Low-ℓ phenomenology closure

7.1 Calibrated physical baseline

At the physical baseline

$$A_{\rm sft} = 0.4378, \qquad \epsilon_\Phi = \frac{1}{4\pi^2}, \qquad c_\chi = 1, \qquad A_{\rm inst} = 1, \qquad \sigma_z = 20, \tag{43}$$

the solver-level source produces a peak low-ℓ correction of approximately

$$\Delta D_\ell \approx +0.34\text{–}0.37\ \mu\mathrm{K}^2 \qquad \text{at} \qquad \ell = 4, \tag{44}$$

with the source confined to roughly $\ell = 2$–8 and no measurable effect on the acoustic peaks or damping tail.

The corresponding low-ℓ penalty is

$$\Delta\chi^2(\ell \le 29) = 2 \times 10^{-6}, \tag{45}$$

placing the physical baseline six orders of magnitude below detectability in this metric and deeply inside the allowed region.

7.2 Amplitude-position map

A focused calibrated exploration was performed over

$$\epsilon_\Phi \in \{0.5, 1, 2, 5\} \times \frac{1}{4\pi^2} \tag{46}$$

and

$$c_\chi \in \{1.0, 0.3, 0.1\}, \tag{47}$$

with $A_{\rm inst} = 1$ and $\sigma_z = 20$ fixed. The key outcomes are listed in Table 2.

Table 2: Focused exploration at the calibrated amplitude.

ϵ_Φ mult	c_χ	peak ℓ	ΔD_ℓ (μK^2)	peak (%)	ISW RMS (%)	full RMS (%)	SFT/CV
0.5×	1.0	4	+0.18	+0.020	0.006	0.001	0.0004
0.5×	0.3	16	+1.94	+0.226	0.126	0.014	0.0092
0.5×	0.1	30	+9.20	+0.872	0.613	0.090	0.0481
1.0×	1.0	4	+0.37	+0.040	0.012	0.001	0.0008
1.0×	0.3	16	+3.89	+0.452	0.253	0.027	0.0184
1.0×	0.1	30	+18.60	+1.761	1.235	0.182	0.0973
2.0×	1.0	4	+0.73	+0.080	0.023	0.003	0.0017
2.0×	0.3	16	+7.80	+0.907	0.507	0.055	0.0368
2.0×	0.1	31	+38.55	+3.596	2.504	0.373	0.2018
5.0×	1.0	4	+1.84	+0.200	0.058	0.006	0.0042
5.0×	0.3	16	+19.67	+2.287	1.276	0.137	0.0929
5.0×	0.1	31	+102.31	+9.543	6.520	0.999	0.5356

The factorization is clean:

- changing ϵ_Φ changes amplitude but not location,
- changing c_χ changes location but not the structural form of the source,
- the baseline remains far below cosmic variance everywhere.

7.3 $A_{\rm inst}$ degeneracy and source-width checks

At fixed calibrated amplitude and $c_\chi = 1$, the $A_{\rm inst}$ sweep gives:

$A_{\rm inst}$	$\Delta\chi^2(\ell \le 29)$	peak ℓ	ΔD_ℓ (μK^2)	peak (%)	active ℓ
1	2×10^{-6}	4	+0.37	+0.040	2–8
10	0.001878	15	+3.59	+0.423	2–24
100	0.090946	30	+18.60	+1.761	2–119

This confirms the solver-level degeneracy predicted by Eq. (39): the response is controlled by $k_c \propto \sqrt{A_{\rm inst}}/c_\chi$.

At fixed calibrated amplitude and band placement, varying the source width gives:

σ_z	$\Delta\chi^2(\ell \leq 29)$	peak ℓ	ΔD_ℓ (μK^2)	peak (%)	SFT/CV
5	0.000000	4	+0.09	+0.010	0.0002
20	0.000002	4	+0.37	+0.040	0.0008
200	0.000325	4	+3.58	+0.390	0.0083

Thus the source width changes amplitude but not location, again exactly as expected.

7.4 Allowed region and closure threshold

The low-ℓ TT fit-quality map can be summarized very compactly:

- 11 of 12 calibrated phase-space points satisfy $\Delta\chi^2(\ell \leq 29) < 1$,
- the only point exceeding unity is the most aggressive explored corner, $\epsilon_\Phi = 5 \times (1/4\pi^2)$ with $c_\chi = 0.1$,
- all runs with $c_\chi \geq 0.3$ preserve the acoustic peaks to machine precision.

A compact allowed-region summary is therefore

$$\{\epsilon_\Phi \leq 5/(4\pi^2),\ c_\chi \geq 0.1\} \qquad \text{at} \qquad \Delta\chi^2(\ell \leq 29) < 1, \tag{48}$$

with the physical baseline in Eq. (43) lying roughly 5×10^5 inside the allowed region in this metric.

V10.1 phenomenology verdict. The shatter source is implemented in CLASS, calibrated against the pre-Boltzmann builder, validated across amplitude / transfer-factor / band / width sweeps, and shown to sit deeply within the Planck-allowed low-ℓ region at the physical baseline. This meets the v1 phenomenology closure threshold.

8 What is closed and what remains open

8.1 Closed enough to preserve

The following statements are now closed enough to state without hedging at the v1 level:

1. the shatter can be written as a field-theoretic instability of the lag field with control variable Λ,
2. the post-critical order parameter obeys $\Phi_{\rm eq} \propto \sqrt{\Lambda - 1}$,
3. the relic field can be inserted into the photon-baryon sector as a real source,
4. the source coefficient β_Q and the k^2/a^2 structure have a first-principles home in the free-energy functional,

5. the instability-band rule in Eq. (39) is solver-real,
6. the calibrated source is phenomenologically allowed and likelihood-mapped at low ℓ.

8.2 Still open

The following remain open and should be treated as the next phase rather than as missing pieces of V10.1 itself:

1. a first-principles microscopic derivation of $f(T)$ from the actual independent-clock carrier sector,
2. a table-driven CLASS mode using fully precomputed $\beta_Q(\tau)$, $\Phi_{\rm eq}(\tau)$, and $k_c(\tau)$ histories,
3. full Planck likelihood work beyond the cosmic-variance penalty metric,
4. broader parameter-estimation and cross-observable exploration.

9 Conclusion

V10.1 closes the early-universe shatter program to the first honest phenomenology threshold. The field-theoretic mechanism,

$$\Lambda \to M_{\rm eff}^2 \to \Phi_{\rm eq}, \tag{49}$$

the transfer chain,

$$\delta\Phi \to \delta Q_\gamma \to \Theta_\ell, \tag{50}$$

and the implementation-level source,

$$S_{\rm SFT}(k,\tau) = A_{\rm sft} g(\tau)\beta_Q(\tau)\frac{k^2}{a(\tau)^2}\Phi_{\rm eq}(\tau)\eta_k W\left(\frac{k}{k_c(\tau)}\right), \tag{51}$$

now form one continuous chain.

The resulting CMB phenomenology is neither hand-waved nor tuned into contradiction. It is present, calibrated, low-ℓ, and comfortably allowed. That is the right outcome for a first extension beyond V10. The next stage is numerical refinement and parameter exploration, not additional symbolic closure.

Final status of V10.1. Field-theoretic shatter mechanism: derived at the current frontier. Minimal Boltzmann bridge: derived and narrowed to its valid limit. β_Q source law and instability band: derived to the current frontier. CLASS implementation and calibration: complete for v1. Low-ℓ phenomenology closure: reached.